

Ibrahim Tayeb

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EDUCATION

University of Illinois - Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- GPA: 3.7/4.0
- Coursework: Computer Architecture, Data Structures, Algorithms, System Programming, Software Design, Applied Machine Learning, Probability & Statistics for Computer Science, Database Systems, Numerical Methods, Cryptography, Linear Algebra, Discrete Structures

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Uber

San Francisco, CA

Robotics Software Engineer

August 2023 – May 2026

University of Illinois - Urbana-Champaign (NASA Lunabotics)

Urbana, IL

- Developed algorithms in C++/Python for autonomous navigation, leveraging ROS2 & SLAM toolbox to enable real-time path planning, resulting in a 25% improvement in movement efficiency and reduced collision incidents
- Integrated LIDAR sensors, computer vision cameras, and motor control units with Raspberry Pi microcomputers, achieving enhanced spatial awareness and precision control, improving positioning accuracy by 30%
- Conducted iterative testing and troubleshooting, utilizing simulations with Gazebo, reducing system errors by 20%

Software Engineer Intern

May 2025 – Aug. 2025

Amway

Grand Rapids, MI

- Architected scalable AWS infrastructure, containerizing applications with Docker & deploying to ECS Fargate
- Automated CI/CD pipelines using CodePipeline, achieving 99.9% uptime and reducing deployment time by 75%
- Built Streamlit dashboard with API integration, session state management, & interactive UI for data visualization & management, resulting in 40% faster data analysis workflows, & enabling users to access insights 5x faster
- Designed modular service architecture consuming REST APIs with OAuth2 authentication, token caching, error handling, and data transformation layers for seamless backend integration, improving system reliability by 60%
- Leveraged Claude Code for automated code generation, & engineered context-aware prompts to accelerate development speed by 50% through improved code quality and consistency

Software Engineer Intern

May 2024 – Aug. 2024

Amway

Grand Rapids, MI

- Designed and developed an internal full-stack web application using ASP.NET, HTML, CSS, JavaScript (frontend), and C#, SQL Server (backend) to streamline the process of retrieving shipment pallet stacking schematics
- Leveraged MS SQL Server to handle data storage and retrieval, allowing engineers to efficiently filter results based on specific criteria, improving decision-making and reducing shipment preparation time by 30%
- Collaborated closely with packaging engineers to ensure the application was user-friendly & met their requirements

PROJECTS

AlphaFlow Execution Engine | *Pandas, NumPy, Scikit-learn, XGBoost, TA-Lib, Plotly, Streamlit, Polygon.io API*

- Designed sophisticated backtesting engine with realistic portfolio management, commission modeling, and slippage simulation to evaluate strategy performance across multiple timeframes and risk metrics
- Developed multi-strategy execution simulator supporting Market Orders, Limit Orders, TWAP, and VWAP execution with market impact modeling and smart order routing across trading venues
- Applied advanced machine learning techniques for financial time series prediction, implementing feature selection, cross-validation, and model performance evaluation with confusion matrices and feature importance analysis

TECHNICAL SKILLS

Languages: Python, C++, Java, C#, JavaScript, HTML/CSS, SQL

Frameworks: React, Express, Angular, Node.js, Flask, AWS cdk, .NET, ROS2

Developer Tools: Git, GitHub, AWS, Docker/Podman, Bitbucket, MongoDB, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, MS SQL Server, Linux (Ubuntu 22.04), Postman, Jupyter Notebook, Elasticsearch

Libraries: Pandas, NumPy, Matplotlib, NLTK, Scikit-Learn, TensorFlow, PyTorch, Streamlit, Requests, Plotly